Assignment No. 7

Name: Vedant Wagh
Roll No: C47
Division: C

Program

```
#include <stdio.h>

#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void addJob()
{
    int job;

    if ((rear + 1) % MAX == front)
    {
        printf("Queue is Full! Cannot accept more jobs.\n");
        return;
    }

    printf("Enter Job Number: ");
    scanf("%d", &job);

    if (front == -1)
    {
        front = 0;
        rear = 0;
    }
    else
    {
        rear = (rear + 1) % MAX;
    }

    queue[rear] = job;

    printf("Job %d added successfully.\n", job);
}

void deleteJob()
{
    if (front == -1)
    {
        printf("Queue is Empty! No job to process.\n");
        return;
    }

    printf("Job %d is processed.\n", queue[front]);

    if (front == rear)
    {
        front = -1;
        rear = -1;
    }
    else
    {
        front = (front + 1) % MAX;
    }
}

void display()
{
    int i;

    if (front == -1)
    {
        printf("Queue is Empty.\n");
        return;
    }

    printf("Jobs in Queue: ");
    i = front;
```

```
    while (1)
    {
        printf("%d ", queue[i]);

        if (i == rear)
            break;

        i = (i + 1) % MAX;
    }

    printf("\n");
}

int main()
{
    int choice;

    do
    {
        printf("\n----- CIRCULAR JOB QUEUE -----\n");
        printf("1. Add Job\n");
        printf("2. Delete Job\n");
        printf("3. Display Queue\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                addJob();
                break;

            case 2:
                deleteJob();
                break;

            case 3:
                display();
                break;

            case 4:
                printf("Program terminated.\n");
                break;

            default:
                printf("Invalid choice!\n");
        }

    } while (choice != 4);

    return 0;
}
```

Output

```
----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
Enter your choice: 1
Enter Job Number: 101
Job 101 added successfully.

----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
Enter your choice: 1
Enter Job Number: 102
Job 102 added successfully.

----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
```

```
Enter your choice: 1
Enter Job Number: 103
Job 103 added successfully.

----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
Enter your choice: 3
Jobs in Queue: 101 102 103

----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
Enter your choice: 2
Job 101 is processed.

----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
Enter your choice: 3
Jobs in Queue: 102 103

----- CIRCULAR JOB QUEUE -----
1. Add Job
2. Delete Job
3. Display Queue
4. Exit
Enter your choice: 4
Program terminated.
```